

JAEHYEON KIM

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rlawogus05



SELF INTRODUCTION

I am a science enthusiast who values curiosity, creativity, and critical thinking. I started with a deep love for pure math, but the desire for practical impact naturally guided me to AI. My research mainly focuses on developing **efficient guidance mechanisms** and **optimization strategies for diffusion models**. Also, I used to enjoy a unique hobby: proving math theorems using physics.

EXPERIENCE

Undergraduate Researcher

AIBL Lab

Mar. 2026 - Present

Seoul National University

EDUCATION

B.S. in Department of Mathematical Sciences

Seoul National University

Mar. 2021 - Present

- Major in Mathematics
- Minor in Computer Science and Engineering
- Minor in Physics
- Completed Mandatory Military Service (May. 2024 - Feb. 2026)

PUBLICATIONS

Journal Articles

- J. Kim, "Hydrostatic-based proofs in geometry," *American Journal of Physics*, vol. 93, no. 8, pp. 682–684, 2025.
- J. Kim, "Water-based proof of jensen's inequality," *The American Mathematical Monthly*, vol. 132, no. 5, pp. 448–452, 2025.

Conference

- J. Kim*, S. Jo, Z. Lee, H. Choi, and K. Kim†, "Compose once, sample many: Amortized composition of diffusion models," Under Review in *ICLR 2027*.
- J. Kim*, Z. Lee*, S. Choi*, and K. Kim†, "Value-informed schedule trimming for accelerated sequential monte carlo guidance in discrete diffusion," Under Review in *ICLR 2027*.
- Z. Lee* et al., "Risk-calibrated proposal transport for finite-particle diffusion steering," Under Review in *ICLR 2027*.